MIL-C-26007C(USAF)
5 April 1967
SUPERSEDING
MIL-C-26007B(USAF)
15 February 1962

MILITARY SPECIFICATION

CHAMBER, DECOMPRESSION, ALTITUDE TRAINING, 20 MAN

1. SCOPE

1.1 This specification covers one type of low pressure high altitude training chamber for indoctrination of flying personnel to the physiological effects of altitudes between sea level and 100,000 feet.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

AA-C-293	Chair, Steel, General Office
SS-T-306	Tile, Floor, Asphalt
TT-E-506	Enamel, Tints and White, Gloss, Interior
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting

Military

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-I-5585	Installation of Low Pressure Oxygen Equipment in Aircraft, General Specification for
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-W-8611	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys; Process for

STANDARDS

Federal

FED-STD-595	Colors (Requirements for Individual Color Chips)

Military

MIL-STD-101	Color Code for Pipelines and for Compressed-Gas Cylinders
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:52
Check the source to verify that this is the current version before use.

MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MS24663	Connector, Plug, Electrical - With Spring-Loaded Pivoted Grounding Blade, Polarized
MS33586	Metals, Definition of Dissimilar

DRAWINGS

Air Force

55B64133	Panel - Observer, Outside
55D64136	Panel - Student Lock and Auxiliary
55D64200	Support Assembly - Chair, Altitude Chamber
55H64118	Panel - Double Student
55H64130	Panel - Inside Observers
55H64168	Desk - Recorders
55H64248	Panel - Lock Operator's
55H64278	Panel - Main Operating
55J64101	Panel - Single Student
55J64126	Panel - Inside Lock Observers
56D64247	Manifold - Oxygen Regulation, Altitude Training Chamber
58D64538	Bracket - Mounting Oxygen Manifold
59H64314	General Panel Configuration-20 Man Hi-Altitude Training Chamber "E" Modification
61E31005	Seat Assy - Second Inside Observer Altitude Chamber "E" Modification
63E31787	Wiring Diagram - AIC-10, 20 Man Altitude

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein: (Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.)

American Society of Mechanical Engineers

Code for Unfired Pressure Vessels

(Copies of the ASME Handbook may be obtained from the American Society of Mechanical Engineers, 29 W 39th Street, New York 18, New York.)

National Fire Protection Association

Pamphlet 70 National Electrical Code

(Copies of the National Electrical Code may be obtained from the National Fire Protection Association, 60 Batterymarch Street, Boston 10, Mass.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

CGS Scientific Corporation

Drawing 10006	Chamber Assembly 20-Man Altitude Chamber
Drawing 10032	Typical Chamber Installation 20-Man Altitude Chamber

(Copies of CGS Drawings may be obtained from the CGS Scientific Corporation, Route 309, Telford, Penna.)

3. REQUIREMENTS

3.1 Components. The chamber shall be built in accordance with CGS Scientific Corporation Drawing 10006. The chamber shall consist of:

Item No	Item or System Name	See Requirements
1	Main and lock compartments	3.6.1
2	Decompression mechanism	3.6.2
3	Evacuation system	3.6.3
4	Oxygen system	3.6.4
5	Air conditioning system	3.6.5
6	Communications system	3.6.6
7	Lighting system	3.6.7
8	Electrical system	3.6.8
9	Controls	3.6.9
10	Control panels	3.6.10

Where conflict exists between the drawings and the specification, the specification shall govern.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Metals. Metals shall be corrosion-resistant steel wherever they are in exposed use, such as shafts, pulleys, hand wheels, etc. Metals which form the structure shall be primed, anodized, parkerized or cadmium plated to resist rust, corrosion or deterioration from fuels, salt spray or atmospheric conditions encountered during storage, shipment or normal service usage.

3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction

3.4.1 Design. The chamber shall be designed to simulate altitude conditions of reduced pressure by varying the pressure within the chamber at controlled rates within the limits of those obtained between sea level and 100,000 feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

3.4.1.1 Safety factor. The main and lock compartments shall be designed using a safety factor of at least 4.

3.4.1.2 Reliability. The chamber shall be designed and constructed as specified herein and to satisfactorily complete the tests specified herein with no failures preventing satisfactory performance. The following procedures and any others deemed necessary by the contractor shall be applied to give a low probability of failure during use: (When specified (see 6.2), a report describing these procedures shall be submitted to the procuring activity.)

(a) Analyze designs selected or considered to determine what failures are most probable and, particularly, what failures are possible that could cause personnel hazards.
(b) Determine what design principles, safety factors, quality control and inspection procedures, and any other procedures not specified herein are necessary.
(c) Determine what use or test data exists for the parts, components and designs selected, or for similar items from which estimates of the expected failure rates can be made.
(d) List the possible failures that could cause personnel hazards leading to fatalities or permanent injury, and at least the ten most probable failures. Determine the engineering reasoning used or to be used in making the design covering those failures.
(e) Determine what tests and inspections, of incoming material or during production, are needed to insure the quality level required to meet the estimates of failure rates.
(f) Estimate and give the engineering reasoning for the estimated number of failures to be expected during the first 2,000 operating hours of the equipment.
(g) Predict the field and maintenance conditions and procedures that should be avoided or cultivated in field use to decrease the probability or frequency of failures. References to periodic maintenance and inspections to be performed may be by reference to technical manuals or other documents prepared for the equipment.

3.4.1.3 Maintainability. The chamber shall be designed and constructed as specified herein to provide the following:

(a) A minimum number of parts consistent with reliability and performance specified herein.
(b) Minimum time and amount of training necessary for assembly, disassembly, location of trouble sources and maintenance, including servicing.
(c) Permit maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools and equipment shall be subject to approval by the procuring activity.
(d) Permit adjustments, servicing, replacements of parts and components and other maintenance with minimum disturbance to other equipment, parts or components.
(e) Permit rapid visual inspection of all parts and components that are most likely to fail, and that could cause personnel hazards if they fail during use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

3.4.2 Construction. The inside and outside walls, the floor and the ceiling of the lock- and main-chamber compartments shall be smooth and continuous. A walkway shall be provided on top of the chamber to prevent damage to the plumbing and wiring. The false ceiling shall be constructed of ten-gage perforated steel panels that can be easily moved for maintenance accessibility. An 0.060-inch-thick sheet-steel skin shall be used to provide a smooth finished outer wall surface. The skin shall be made in removable sections covering plumbing and wiring. Access to any section shall be gained by removal of hex head bolts. Panels shall be identified if plumbing or wiring is covered. In addition, all panels shall have numbers stenciled on each adjacent corner to insure proper matching and bolt-hole alignment. The component parts of the decompression valve, air conditioning inbleed-filter and inlet ducts, electrical junction boxes, oxygen-system connections and related wiring and plumbing shall be installed on top of the chamber. All outer edges shall have commercial 90° finish angle over the skin to prevent personnel cuts. The vacuum pump and compressors shall be mounted in the pump room. Access to the chamber top shall be provided by a ladder constructed of 1-inch OD steel conduit. A 180° curved handrail of similar material shall be welded on each side of the ladder 12 inches above the chamber to serve as a hand hold while ascending or descending. The handrail shall be opposite the lock-operator's control panel as shown on Drawing 59H64314.

3.4.2.1 Floor. The main- and lock-compartment floor shall be covered with asphalt tile conforming to SS-T-306. Suitable baseboard and border shall be provided. The color shall be tan. Particular attention shall be given to cementing the tile to insure that they will not break loose under reduced pressure. A corrosion-resistant steel threshold shall be provided at the door sills. It shall be installed so that it will protect the edge of the tile from breaking or chipping.

3.4.2.2 Miscellaneous. The contractor shall provide and install all valves, fittings, motor starters, junction boxes, connections and other component parts required for proper operation of this chamber when installed in the high-altitude chamber building. The chamber shall be installed in accordance with CGS Scientific Corporation Drawing 10032.

3.4.2.3 Stress relieving. The bulkhead, end sections and doors shall be furnace stress-relieved after fabrication to eliminate the possibility of warping and distortion during use. The stress-relieving procedure shall be in accordance with ASME Code for Unfired Pressure Vessels as prescribed for the material used.

3.4.2.4 Welding. The chamber shall be of welded construction and the welding shall meet the requirements of MIL-W-8611.

3.4.2.5 Piping. All vacuum and air piping shall be of standard steel pipe, flange and welded, or threaded. All threads shall conform to MIL-P-7105. Pipes and ducts shall not be installed inside the chamber main or lock compartments. All pipes and ducts shall be of the same diameter or larger than the corresponding control valves.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

3.4.2.6 Wiring. All wiring shall conform to Pamphlet 70 (see 6.3). All electrical components of the chamber, air conditioner, air compressor and evacuation system shall be protected by resettable self-tripping circuit breakers of adequate amperage. Circuit breakers shall have identifying metal name plates or tags. Code numbers or letters shall not be used to identify the circuit breakers. All circuit controls, electromagnetic starters and related switches for the evacuation pump, air conditioner and decompression valve actuators shall be mounted on the wall inside the pump room. All circuit controls, electromagnetic starters and related switches for major mechanical equipment which present a hazard to personnel when in operation shall be located within sight and, if possible, within arms reach of the actuated equipment.

3.5 Performance.

3.5.1 Evacuation. The evacuation equipment shall decrease the chamber (door between lock and main compartment open) internal pressure from atmospheric pressure at sea level to 8.0 mm Hg pressure. The maximum time for this evacuation shall be 16 minutes.

3.5.2 Main-and lock-compartment leakage. The total leakage of the lock- and main compartments shall not cause the pressure inside the chamber to increase more than 36.7 mm Hg in the first 3 hours and not more than 262 mm Hg over the 24-hour period.

3.5.3 Decompression. Actuation of the decompression valve when in the full-open position (100 percent) shall allow a 7½ psi pressure differential between the lock and main compartments to equalize in not more than 500 milliseconds when tested in accordance with 4.5.4.

3.5.4 Oxygen system. The oxygen line leakage rate shall not reduce the 450 psi line pressure more than 100 psi over a 24-hour period.

3.5.5 Air conditioning system. The air conditioning system shall be capable of maintaining $70^{\circ} \pm 3^{\circ}$ F with a relative humidity of not greater than 50 percent inside the chamber.

3.5.6 Lighting system. With all the lights on, the intensity shall be 50 foot-candles when measured with a light meter held 3 feet above the floor.

3.5.7 Controls. All controls and valves shall perform their desired functions without requiring excessive force.

3.5.8 Noise level. Noise level inside and outside the chamber shall not exceed the values specified in 3.6.3.

3.5.9 Communication system. The communication system shall operate at all altitudes without excessive interference.

3.6 Details of components.

3.6.1 Main and lock compartments.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T07:45Z
Check the source to verify that this is the current version before use.

3.6.1.1 Size. The chamber shall be rectangular. It shall be divided into a main compartment and a lock compartment by means of a fixed bulkhead. The main compartment shall be 20 feet long by 8 feet wide. The lock compartment shall be 6 feet long by 8 feet wide. Both compartments shall be 7 feet high from the floor to the false ceiling. The outside dimensions shall not exceed 27 feet 4 inches in length, 9 feet 4 inches in width (without control panels installed) or 11 feet to the top of the decompression control valve or any related plumbing and component parts. The lock compartment shall provide facilities for eight students and one inside observer. The main compartment shall provide facilities for sixteen students and two inside observers.

3.6.1.2 False ceiling. The area above the false ceiling shall require 12 to 18 inches for the air-conditioning system.

3.6.1.3 Doors. An outwardly opening door 6 feet in height by 2-1/2 feet in width shall be installed to provide entry into the lock compartment. An identical door in the lock compartment shall provide entry into the main compartment. The door shall swing open into the lock from the bulkhead separating the lock from the main compartment, as shown on Drawing 59H64314. The doors shall be supported by bearings at all hinges. Each door shall seal against the frame in the bulkhead by means of a rubber gasket. The doors shall seal initially by means of four cam-type closures that fall clear as the internal chamber pressure is reduced. Doors shall be so installed as to fall open at least 4 inches when the cam locks are not engaged and pressure differential is nonexistent. An oxygen-safe nonpetroleum base lubricant shall be used on bearings. The cam lock mounting studs shall be not less than 7/8 inch in diameter. Provisions to compensate for wear of the locking surfaces shall be made through the use of 1/16-inch steel washers installed on both sides of the cam lock. The cam lock shall be manufactured from solid round steel bar stock not less than 1-1/4 inches in diameter. A rubber strike pad shall be installed on the doors to absorb the noise usually made when the cam locks are released during evacuation.

3.6.1.4 Windows. The following observation windows shall be provided:

Number	Size	Location
1	18 by 86 inches	In the end bulkhead of the main compartment opposite the lock
6	18 by 34 inches	Two on each side of the main compartment and one on each side of the lock
2	18 by 18 inches	One in each personnel entry door

The bottom of each window shall be approximately 50 inches above the floor. All windows shall be of double-pane temper glass (Tuflex or equal). The outer panes shall be a minimum of 1 inch thick, set in rubber gaskets of "ell" cross section and shall be molded in one piece in accordance with the glass dimensions. The inner glass shall consist of two pieces of 3/4-inch tempered plate, laminated together with a 1/8-inch vinyl inner layer. The inner panes shall set in rubber gaskets of "ell" cross section and shall be molded in one piece in accordance with the glass dimensions. The air space between the multipanes shall be

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:45Z
Check the source to verify that this is the current version before use.

protected from accumulation of moisture by the use of Silica-Gel crystals conforming to MIL-D-3464. The crystals shall be contained in perforated transparent plastic tubing 1/2-inch OD and 10 inches long, sealed on each end. Between the outer and inner panes there shall be a flat rubber gasket molded in one piece in accordance with the glass dimensions. Putty or sealing compounds shall not be used. Glass shall be designed to withstand the extreme pressure and pressure changes. Window frames shall be of continuous construction extending around all four sides of the window.

3.6.1.5 Access plugs. Ten access plugs shall be provided. The access plugs shall consist of 1-inch pipe sections with a plug in each end. The access plugs shall be approximately 12 inches above the floor and approximately equally spaced around the chamber except for the doors.

3.6.1.6 Seats. The seats shall be of the bench type in smoothly finished 2- by 8-inch oak planks to provide a 20-inch width. The backrest shall be similar. The bench and back shall be supported on 1-1/2-inch wide steel straps. Cushions shall be attached to the bench between the student consoles and in the main and lock compartments. A seat shall be attached to the inside of the main compartment entry door for use by a second inside observer. The seat shall conform to Drawing 61E31005.

3.6.2 Decompression mechanism. Explosive or rapid decompression shall be accomplished by the use of a single or multiple balance air-type decompression valve. Means shall be provided to vary and limit the amount of travel of the valve disc from full-open to full-closed position. A fixed indicator shall be provided to show the amount of travel. The balanced-air design of the valve shall be such as to enable a constant rate of valve travel, independent of the pressure differential across the valve. The valve disc shall seat tightly against the face of the valve body flange and there shall be no leakage through the valve at any differential pressure between 0 and 15 psi. A butterfly valve shall be provided between the decompression valve and the main compartment to eliminate personnel hazard through inadvertent decompression. The operating control shall be mounted near the lock control panel. A position indicator lamp shall be placed adjacent to the decompression valve safety switch on the lock operator's control panel. The decompression mechanism shall be arranged as shown on Drawing 59H64314.

3.6.3 Evacuation system. Altitude simulation shall be accomplished through the use of a rotary nonpulsating-type vacuum pump with a minimum capacity of 700 cfm at normal rated rpm. Lubrication shall be accomplished by use of tricresyl phosphate to minimize the danger of fire or explosion present through the mixture of oxygen with oil. The vacuum pumps shall be furnished complete with motor starters and related electrical equipment to be operated on a 440V, 3-phase 60-cycle alternating current. The pump, motor base, coupling, lubricant separator, belt guard, vacuum reservoir, necessary solenoid valves and any other equipment used in the evacuation system shall be supplied with a tricresyl-phosphate-resistant finish paint. The pipe lines interconnecting the system shall be of welded and flanged construction. The pump and header arrangement shall be as shown on Figure 1. Flexible steel expansion joints shall be supplied to absorb vibration between pump and vacuum line into the chamber. To minimize the noise level inside and outside the chamber, all vacuum and inbleed

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

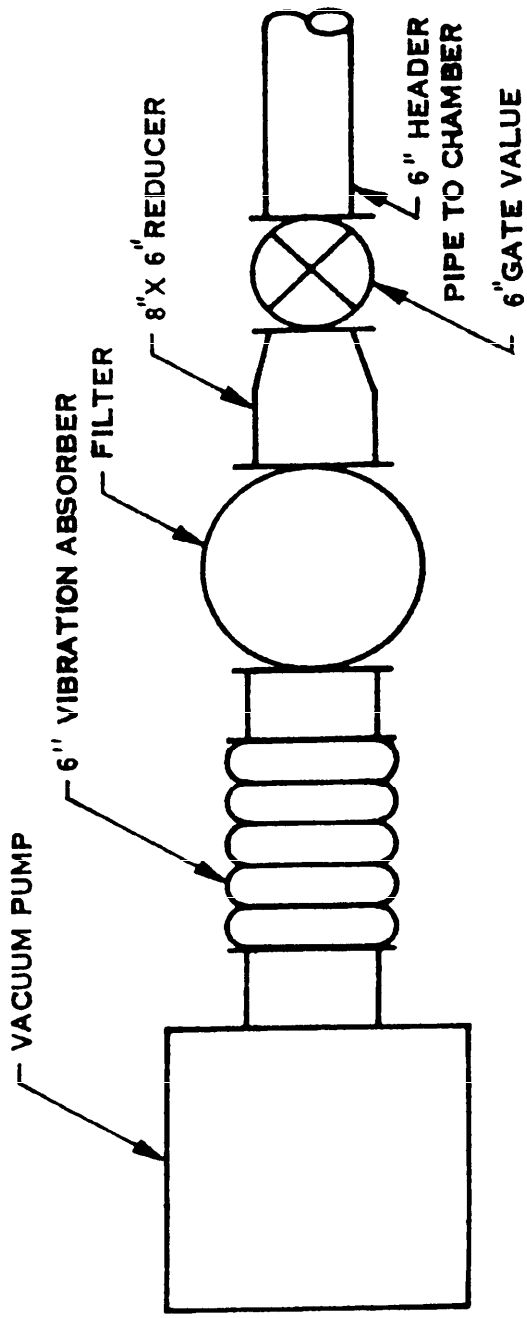


FIG. 1 Pump and Header Arrangement

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

air lines shall be provided with suitably-sized silencers. The inbleed air lines shall enter into the main and lock compartments of the chamber at a point 12 inches above the chamber floor. The silencers shall be capable of withstanding a pressure differential equivalent to a full vacuum. The silencers shall be mounted on the roof of the chamber and shall reduce the entry air noise level below the following values:

Frequency Per Second	Noise Level in Decibels
20-75	88
75-150	78
150-300	70
300-600	64
600-1,200	58
1,200-2,400	54
2,400-4,800	52
4,800-10,000	51

3.6.3.1 Pump pad. The contractor shall provide at the installation site a reinforced concrete pad 20 inches thick to support the 700 cfm evacuation pump and component parts. The pad shall be isolated from vibration.

3.6.3.2 Wire screens. Wire screens shall be installed over the intake of the vacuum lines. The mesh shall be of sufficient size to prevent ice accumulation on the mesh when the vacuum line valve is in the maximum open position and a vacuum is being developed. A copper- or brass-wire screen (1/16 inch mesh hardware cloth) shall be required at the connection between the flange of the vacuum pump inlet flange and the vacuum pipe flange to prevent foreign body damage to the pump.

3.6.4 Oxygen system. The oxygen system shall be composed of an oxygen control panel conforming to Drawing 56D64247 and student and observer's stations for oxygen regulators. The oxygen pressure shall be reduced from 2,000 to 450 psi by the pressure-reducing valves on the oxygen control panel. The panel shall utilize mounting brackets conforming to Drawing 58D64538 plus tubing, nuts, bolts and washers to completely install the oxygen control panel in the oxygen supply room. Six high-pressure flexible hoses shall be provided to connect the oxygen supply bottles to the panel inlet blocks. All the oxygen regulators in the chamber shall be connected to one main supply line on the chamber. This supply line shall be connected to all the oxygen pressure gages on the main and lock operator's panels. The main supply line shall exit from the chamber in such a manner that the oxygen supply line from the oxygen room can be connected to it. Reference student and inside observer's panels for location of oxygen stations.

3.6.4.1 All fittings, thread compound, tubing and accessories used in the oxygen system shall be of a type suitable for use with 100 percent oxygen under pressure and shall be compatible with oxygen equipment listed in MIL-I-5585. All oxygen system components shall be maintained in a state of cleanliness or cleaned in a manner that renders them suitable for use with 100 percent oxygen under pressure. A one-cylinder emergency oxygen manifold shall be installed adjacent to the lock-operator's panel and connecting to the oxygen delivery line close to the point of chamber penetration.

Source: https://assist.dla.mil -- Downloaded: 20260-20T07:45Z
Check the source to verify that this is the current version before use.

3.6.5 Air conditioning system. The main and lock compartments shall be provided with ceiling-mounted cooling units located in the space above the false ceiling. These units shall be mounted flush with the false ceiling. The cool air shall be directed along the false ceiling and not directly down on the students. The air-conditioning unit shall use a freon-type refrigerant and a water-cooled condenser. The condenser shall use not more than 6 gallons of water per minute. Each cooler shall be equipped with a blower capable of circulating the air in the chamber. A cooling unit shall also be installed in the inbleed air intake and located externally on top of the chamber. The inbleed air cooler shall be provided with washable air filters capable of removing 5 micron dust particles. The humidity and temperature in the lock and main compartments shall be individually controlled by thermostats located in each compartment. The minimum range of the thermostats shall be from 50° to 90° F. The units shall be capable of operating on either 110 or 120V 60-cycle electric system. The compressors and condenser shall be installed in the pump room. The air-conditioning, cooling and condensing units shall be approved by Underwriters' Laboratories, Inc. The air blower motors shall be capable of operation to a simulated altitude of 100,000 feet. Provisions shall be made for draining condensation from each cooling unit to the outside into individual accumulator tanks provided with shutoff valves to prevent reverse flow during chamber operation.

3.6.6 Communications system. The communications system shall be composed of AIC-10A intercommunication components including amplifiers and loudspeaker. Provision for utilization of a 24V battery as an emergency power source shall be made in the installed wiring harness in accordance with the wiring diagram shown on Drawing 63E31787.

3.6.7 Lighting system. The chamber shall be illuminated by fluorescent lamps that shall give a minimum of 50 foot-candles evenly distributed 3 feet above the chamber floor. The ballasts shall be capable of withstanding evacuation above 100,000 feet and explosive decompression from 80,000 feet without failure. Lamps and related fixtures shall withstand extreme and rapid pressure changes without failure. The lamps provided shall be of the daylight type. The lighting fixtures shall be mounted flush with, or above, the false ceiling, have covered tubes and provide easy access for maintenance or replacement. An emergency lighting system utilizing 24V dc power shall be provided by flush-type incandescent lamps (Luma-line or equal) and related fixtures.

3.6.8 Electrical system. The main chamber compartment shall have four standard 110V double-outlet receptacles, two on each side. The lock compartment shall have two standard 110V double-outlet receptacles, one on each side. Identical receptacles shall be provided in approximately the same location on the outside walls. The boxes shall be located approximately 12 inches above the floor. Electrical receptacles and plugs shall conform to MS24663.

3.6.9 Controls. All control valves shall be quick-opening type and capable of controlling the flow of air between their full-open and closed positions. All data and handbooks shall refer to descent control valves as "air controls" and climb controls as "vacuum controls." Both the lock and main compartment shall have vacuum and inbleed air valves. The chamber shall be provided with both

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

fine and coarse flow control valves. The fine control valves shall bypass the course valves and be operated from the control handles in the operator's panels. The linkage between the control handle and the valve shall be a straight rod with a fork-type connecting link to prevent eccentric loading on the linkage. The open position for the valves shall be when the control handle is pulled toward the operator. The coarse control valves shall be located beneath the operator's control panel and open when operated in a downward direction.

3.6.9.1 Vacuum valves. The main compartment shall be provided with a 4-inch ID coarse control valve and a 2-1/2-inch ID fine bypass valve. The valves shall control the flow of air out of the main compartment. The lock compartment shall be provided with a 3-inch ID coarse control valve and a 2-inch ID fine bypass valve. These valves shall control the flow of air out of the lock compartment. All vacuum valves shall be so located that the operator can control them with his left hand when in a normal position.

3.6.9.2 Air valves. The coarse and fine inbleed air valves shall be the same size as their respective vacuum valves. The main compartment valves shall control the flow of air into the main compartment from the inbleed cooler. The lock compartment valves shall control the flow of air into the lock compartment from the inbleed cooler. All air control valves shall be so located that the operator can control them with his right hand when in a normal position.

3.6.9.3 Equalizer valves. All equalizer valves shall be a minimum of 3 inches ID and painted red. All valves shall open with a downward motion. One equalizer valve shall be located above the lock observer's window within easy reach of the lock operator when sitting at his control panel. Two additional valves shall be installed in the bulkhead between the lock and main compartments. One valve shall be controlled from inside the lock compartment and the other from inside the main compartment. The pressure between the two compartments shall be equalized when these valves are operated. Equalizer valves shall be located on each end of the chamber and operated from inside. They shall equalize the pressure between the inside and outside of the chamber. A red decal shall be placed near all equalizer valves in a position where it can be easily read. The following information shall be printed on each decal in 3/8-inch white letters:

WARNING: USE IN EXTREME EMERGENCY ONLY.

3.6.9.4 Rapid decompression assembly. The decompression valve shall operate remotely from the lock operator's control panel. A safety switch shall be provided to prevent inadvertent actuation of the rapid decompression valve. A 5 cfm air compressor, plus lines, trap and filter, solenoid actuators and related electrical equipment shall be provided for the control and operation of the decompression valve.

3.6.10 Control panels. The lock and main compartment shall be provided with control panels containing instruments, intercommunication equipment, switches and controls to maintain their respective compartments at the desired altitude and to conduct decompression simulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

3.6.10.1 Main control panel. The main control panel, Drawing 55H64278, shall be located as shown on Drawing 59H64314. All items shall be identified with nameplates. The following items shall be installed on the chamber within arm's reach of the operator while he is sitting in his normal position:

(a) Main compartment coarse vacuum control valves.
(b) Main compartment coarse air control valves.
(c) Air-conditioning shut-off switch.
(d) Air compressor shut-off switch.
(e) Chamber light switch.
(f) Manometer, Type FA-129, Range 0 to 800 mm Hg, calibration 0 to 800 mm Hg and -1000 to +200,000 feet altitude, Wallace and Tiernan Inc manometer or equal.
(g) Fine air and vacuum control valves shall be operated by handles on the panel.

3.6.10.2 Lock control panel. The lock control panel, Drawing 55H64248, shall be located as shown on Drawing 59H64314. All items shall be identified with nameplates. The following items shall be installed on the chamber within arm's reach of the operator while he is sitting in his normal position:

(a) Lock compartment coarse vacuum control valves.
(b) Lock compartment coarse air control valves.
(c) Lock compartment light switch.
(d) Equalizer valve handle.
(e) Butterfly valve operating control.
(f) Manometer, Type FA-129, range 0 to 800 mm Hg, calibration 0 to 800 mm Hg and -1000 to +200,000 feet altitude, Wallace and Tiernan Inc or equal.
(g) Butterfly valve position indicator lamp.
(h) Fine air and vacuum control valves shall be operated by handles on the panel.

3.6.10.3 Observer panels. The chamber shall be provided with five outside observer panels conforming to Drawing 55B64133. The panels shall be located above the windows as shown on Drawing 59H64314. An observer panel conforming to Drawing 55H64130 shall be located inside the chamber in the main compartment above the window in the end bulkhead. Another observer panel conforming to Drawing 55H64130 shall be mounted in the ceiling above the entrance door similar to the first inside observer panel. The lock compartment shall be provided with an observer panel conforming to Drawing 55J64126 and located in the bulkhead between the lock and main compartments as shown on Drawing 59H64314. One recorder's desk conforming to Drawing 55H64168 shall be provided and shall be installed in accordance with Drawing 63E31787.

3.6.10.4 Student panels. The chamber shall be provided with six double student panels conforming to Drawing 55H64118 and four single student panels conforming to Drawing 55J64101. The lock compartment shall be provided with eight student panels conforming to Drawing 55D64136. All student panels shall be located as shown on Drawing 59H64314.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

3.6.11 Operator's and observer's chairs. Operator's and outside observer's positions shall be equipped with seat support brackets conforming to Drawing 55D64200 and chairs conforming to AA-C-293, type II, class 2, style B. The chairs shall not be required to have the base called out in AA-C-293 since they will be supported by the seat brackets attached to the chamber. The chair brackets shall be attached to the chamber in a manner which will provide adequate seating and allow the operators and observers to perform their desired functions and permit them to have a good inside view of the chamber. The brackets shall be located as shown on Drawing 59H64314.

3.7 Painting and finishing. All accessible carbon steel surfaces shall be thoroughly cleaned of dirt, rust, scale, oil, grease or other matter. The chamber shall be primed with two coats of red oxide primer conforming to TT-P-664 before shipping from the contractor's plant. After completion of the installation at the Air Force chamber site, the interior and exterior of the chamber and all related plumbing shall be reprimed after all welding has been completed. The interior of the chamber shall be painted light green with low-luster enamel, Pittsburg Paint 20-6, Color 32P11 or equal. The exterior of the chamber shall be painted light blue with low-luster enamel, Pittsburg Paint 20-6, Color 5P31 or equal. The door latches, foot rests, ladders and chair-support brackets shall be painted gray, Color 16376, conforming to FED-STD-595. Painting shall be in accordance with TT-E-506. This does not apply to the control panels, vacuum and air lines or cadmium or chromium plating. The vacuum supply line pipes shall be painted aluminum as the primary color. Unless otherwise specified, all pipelines and valves shall be painted in accordance with MIL-STD-101. All components of the evacuation system in the pump room shall be painted with a tricresyl-phosphate-resistant paint in the color scale closest to Color 16376 of FED-STD-595.

3.8 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.8.1 Marking. All controls, instruments and switches on the control panel or as otherwise located shall be marked by engraved plates to identify or indicate its function. Code letters or numbers shall not be used in place of the actual name of the item.

3.8.1.1 Serial numbers. Serial numbers shall be assigned to all items as specified by the procuring activity.

3.9 Workmanship. The chamber shall be manufactured in a thoroughly workmanlike manner. The chamber shall be clean and free of any defects that might affect appearance, functionability, maintainability and the safety of the student or operating personnel. All details not specifically covered herein shall be of such high quality that the chamber will be suitable for continuous use in all respects for an indefinite period of time.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the decompression chamber shall be classified as acceptance tests (see 4.4). These tests shall be conducted at the designated Air Force Base after installation of the chamber facility.

4.3 Test conditions.

4.3.1 Atmospheric conditions. Unless otherwise specified, all tests shall be made at an atmospheric pressure between 28 and 32 inches Hg at a temperature of $77^{\circ} \pm 18^{\circ}$ F and a relative humidity of 80 percent or less. Where tests are made when the temperature and atmospheric pressure are substantially different from the above values, proper allowance shall be made for the change in instrument readings.

4.4 Acceptance tests. Acceptance tests shall consist of the individual tests.

4.4.1 Individual tests. Each chamber shall be subjected to all the tests described under 4.5.

4.4.2 Test report. After the contractor completes the acceptance tests, he shall prepare a test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity (see 6.2).

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the above report (same number and type of copies) or shall be included as part of that report:

(a) All failures, maintenance and other events recorded shall be identified by accumulated operating time, miles, cycles or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.
(b) Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.
(c) Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and reasoning behind the predicted effectiveness of those corrections.
(d) Test activity or contractor comments on item features or requirements that if modified should improve the item.
(e) Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the item's reliability and useful life.

Source: https://assist.dla.mil -- Downloaded: 202640-20T07:45Z
Check the source to verify that this is the currentversion before use.

(f) Estimates (± 5 percent of actual experienced) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

4.4.2.2 After receipt of the report, all tests shall be repeated under surveillance of the Air Force representative.

4.5 Test methods.

4.5.1 Examination of product. The chamber shall be inspected to determine compliance with the dimensions, construction, arrangement, workmanship and materials specified herein.

4.5.2 Evacuation. The chamber shall be evacuated to 8 mm Hg pressure in 16 minutes. The evacuation shall be accomplished under the following conditions:

(a) All entrance doors into altitude chamber shall be shut and sealed against the door gasket by means of the cam-type closures.
(b) The door between the lock and the main compartments shall be left open.
(c) The air-conditioning system shall be turned off.
(d) All inbleed air valves and equalizer valves shall be in the closed position.
(e) All oxygen regulators inside the chamber shall be in the off position.
(f) Vacuum pump shall be operating at its rated rpm.

4.5.3 Chamber leakage rate. The leakage rate of the chamber shall not exceed the specified rate in 3.5.2 when tested under the following conditions:

(a) With the door between the main and lock compartments open, the chamber shall be evacuated to 8 mm Hg pressure.
(b) All valves shall be placed in the closed position. Plugs or caps shall not be used on the equalizer valves to prevent them from leaking.
(c) The lock compartment of the chamber shall be evacuated until a 7.5 psi pressure differential exists between the main and lock compartments.
(d) The vacuum pump shall be turned off.
(e) Pressure reading shall be taken hourly over a 24-hour period.

4.5.4 Rapid decompression. Rapid decompression time shall be obtained in the following manner:

(a) All doors shall be sealed against their gaskets by means of the cam-type closures.
(b) The main compartment shall be evacuated to a pressure of 6.7 psi or less.
(c) The lock compartment shall be evacuated until a pressure of 7.5 psi greater than the main compartment pressure is obtained.
(d) The air valves to all the instruments shall be shut off to prevent damage to the instruments.
(e) A pressure recorder shall be attached to a pressure transducer located in the lock compartment. The recorder shall plot the pressure change as a function of time.
(f) The vacuum pump shall be turned off.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

(g) The rapid decompression switch shall be armed with the safety switch.
(h) The recorder shall be started and the RD switch fired.
(i) The test shall be conducted with the RD valve 100 percent, 75 percent, 50 percent and 25 percent open. The charts and graphs obtained from this test shall be included with the results of the test.

4.5.5 Oxygen system. Oxygen supply lines shall be checked for leakage over a 24-hour period. A line pressure of 450 psi shall be obtained from the high pressure oxygen cylinders by reducing the pressure through the oxygen control panel regulators. The oxygen regulators in the student panels and oxygen cylinder valves shall be shut off and pressure readings taken at the beginning and end of the test.

4.5.6 Air-conditioning system. The air-conditioning system shall be tested in the following manner:
(a) Wet and dry bulb temperature readings shall be recorded for each compartment every 3 minutes for a 30-minute period under the following conditions:
(1) Lock and main compartment doors shut.
(2) Chamber coolers operating.
(3) Vacuum pump operating.
(4) Inbleed air valves open.
(5) Vacuum valves open.
(6) Eight men in the main compartment.
(7) Four men in the lock compartment.
(b) Dry bulb temperature readings shall be recorded under the following conditions:
(1) The lock and main compartment door closed.
(2) Descending the lock compartment from a simulated pressure altitude of 100,000 feet at a rate of 3,000 fpm.
(3) Inbleed cooler operating and chamber coolers operating.
(4) Descend the main compartment from a simulated pressure altitude of 100,000 feet at a rate of 3,000 fpm.
(c) The temperature and humidity of the chamber installation room shall be recorded.

4.5.7 Lighting system. All light switches and related components shall be capable of meeting the requirements of 3.5.6.

4.5.8 Controls. All controls and valves shall be operated under conditions of actual use to determine that they satisfactorily perform their desired functions without excessive force.

4.5.9 Noise level. Noise level readings shall be taken under the following conditions with a decibel indicating meter:
(a) Inside the chamber at an altitude of 10,000 feet with all equipment operating and maximum inbleed airflow into chamber.
(b) Outside the chamber at all operators' positions with maximum vacuum and inbleed airflow.

4.5.10 Communications system. The system shall operate at all altitudes without excessive interference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Assembly sketch. A comprehensive sketch of the complete chamber and all components shall be made locating and identifying all component parts and their relation to each other, including all pipe, electrical or mechanical connections. Each separate part shall be identified with a number indicated on the sketch. The sketch shall be placed in an envelope secured to the interior wall of the chamber.

5.2 Disassembly. All major removable components may be removed from the chamber for shipment. Each part shall be numbered by painting or tagging to correspond to the assembly sketch.

5.2.1 All oxygen and air-conditioning pipes and openings shall be capped with masking tape when the chamber is disassembled. Openings in the pumps, air-conditioning compressors and chamber shall be sealed to exclude the entry of dirt or other foreign matter during shipment. All pipes shall be packed and securely fixed in place inside the chamber.

5.3 Loose component parts. The oxygen manifold and other loose component parts shall be boxed or crated and stored inside the chamber where possible.

5.4 Permanently installed equipment. Permanently installed equipment, such as lighting fixtures, shall be left intact.

5.5 Packaging. The chamber and components shall be packed in accordance with good commercial practice. Windows shall be covered and chamber doors shall be securely wired shut. Any required marking shall be in accordance with MIL-STD-129.

5.6 Installation. The contractor shall reassemble all components of the altitude chamber and install it ready for use at the location designated in the contract. All welded vacuum lines shall be thoroughly cleaned of welding slag prior to final attachment to the vacuum pump.

6. NOTES

6.1 Intended use. The altitude chamber is intended to be used for altitude indoctrination of flight personnel in the use of standard oxygen and other protective equipment.

6.2 Ordering data. Procurement documents should specify the following:

(A) Title, number and date of the specification.
(B) The facility where the chamber is to be installed plus a building layout at the facility.
(C) Information to be provided the contractor concerning the building condition, type and service facilities available at the designated Air Force Base.
(D) When reliability report is desired (see 3.4.1.2). Report to be submitted at least sixty days prior to scheduled delivery of the item.

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:45Z
Check the source to verify that this is the current version before use.

(E) That the test report (see 4.4.2) be submitted to the designated activity fourteen days prior to notice by contractor of readiness to perform contract technical compliance inspection.

Fire and casualty hazards. The bidder will submit proof to the procuring ivity that the wiring specified in 3.4.2.6 which he proposes to supply under s specification conforms to the standards of the Underwriters' Laboratories, as regards fire and casualty hazards. The label or listing of the Under- ters' Laboratories, Inc will be accepted as evidence of conforming to the cification requirements.
.1 In lieu of the label or listing, the bidder may submit independent of satisfactory to the procuring activity that this wiring conforms to the lished standards, including methods of tests, of the Underwriters' Labora- ies, Inc. Compliance with the fire and casualty hazards requirements does absolve the bidder from complete compliance with the other requirements this specification.

Asterisks. Asterisks are not used in this revision to identify changes n respect to the previous issue due to the extensiveness of the changes.

todian:
ir Force - 70

Preparing activity
Air Force - 70

Project No 6930-F051

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

OGDEN AIR MATERIEL AREA
ATTN: OONEO
HILL AFB, UTAH 84401

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:45Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

Source: https://assist.dla.mil -- Downloaded: 202610-20T07:45Z
Check the source to verify that this is the current version before use.